

Correlation

When one variable changes, does the other variable also change?

Homes studied	Marks
1	40
2	50
3	60
4	70
5	80

positive correlation

Speed	Time to reach destination
20 km/h	60 min
40 km/h	30 min
60 km/h	20 min

higher speed $\rightarrow$ lesser time
negative relationship

Direction, Strength

r

-1 1

$+1.0 \rightarrow$ perfect positive
 0.8 to $0.99 \rightarrow$ very strong positive
 0.5 to $0.79 \rightarrow$ moderate positive
 0.3 to $0.49 \rightarrow$ weak positive
 0.29 to $0.29 \rightarrow$ almost no relationship
 0.3 to $0.49 \rightarrow$ weak -ve
 0.5 to $0.79 \rightarrow$ moderate -ve
 0.8 to $0.99 \rightarrow$ very strong -ve
 $-1 \rightarrow$ perfect -ve

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Spearman correlation

Strength
direction
 ρ +ve or -ve
non-linear

$\rho \rightarrow$ how closely do points follow a straight line.
 $S \rightarrow$ Do higher values of X generally correspond to higher values of Y.

10000 1	200	300
20000 2	500	400
30000 3	900	600
40000 4	1500	1500
50000 5	3000	1500
500000000	1000000	

distinct relationship in correlation

ρ value

ρ High ρ
 ρ Reliable
 ρ Low ρ

lower ρ value $\rightarrow$ weak rel.
 higher ρ value $\rightarrow$ strong rel.

birds $\rightarrow$ Age 8.
 start.
 stay or not.
 Can engine $\rightarrow$ has die.

Kendall's tau

agreement

association concordant & discordant pairs

Seq.	agrees on ordering	Tau			
A	1	1	5		
B	2	2	3		
C	3	3	1		

Population Sample

entire dataset
 μ, σ, var
 population $n = 50000$
 sample mean $(\bar{x})$
 sample std dev $= s$
 100 data
 $\bar{x} = 48500$

pop (population) $\rightarrow [10, 20, 30, 40, 50]$

popul. $\mu = 30$

$S_1 \rightarrow [10, 20]$
 $\text{mean} = 15$

15, 25, 45, 50 ...

$S_2 \rightarrow [20, 30]$
 $\text{mean} = 25$

sampling distribution.

$S_3 \rightarrow [40, 50]$
 $\text{mean} = 45$

$S_4 \rightarrow [50, 50]$ $\text{mean} = 50$

Central Limit Theorem

distribution of sample means $\rightarrow$ normal distribution.

population
 $C \cdot \sqrt{n}$
 $\mu \bar{x} = \mu$

sample
 Bell curve

Hypothesis testing

AOV
 average order value $= \$1000$

S_1 Null hypothesis (H_0)
 current belief $A = 1000$

Alternative hypothesis (H_1)
 $A \neq 1000$

p value $p = 0.03$
 H_0 3% of time
 $W = 5 \cdot 10^{-6}$
 $W = 7 \cdot 10^{-6}$
 $\rightarrow$ p value $= 0.02$
 $0.02 < 0.05$
 reject H_0

$\Rightarrow$ Absence of evidence is not evidence of absence.

Type I error

H_0 is true
 but we reject it
 false alarm.

Type II error

H_0 is false
 but we fail to reject
 missed detection.

Reality	Decision	Result
H_0 True	Reject H_0	Type I error
H_0 True	Fail to reject H_0	correct
H_0 False	Reject H_0	correct
H_0 False	Fail to reject H_0	Type II error

S_1 State hypothesis H_0 & H_1

S_2 $\alpha = 0.05$

S_3 collect sample data

S_4 test statistics
 $\rightarrow$ Z-score
 $\rightarrow$ T-score
 $\rightarrow$ Chi-square statistics.

S_5 p value

S_6 make decision.